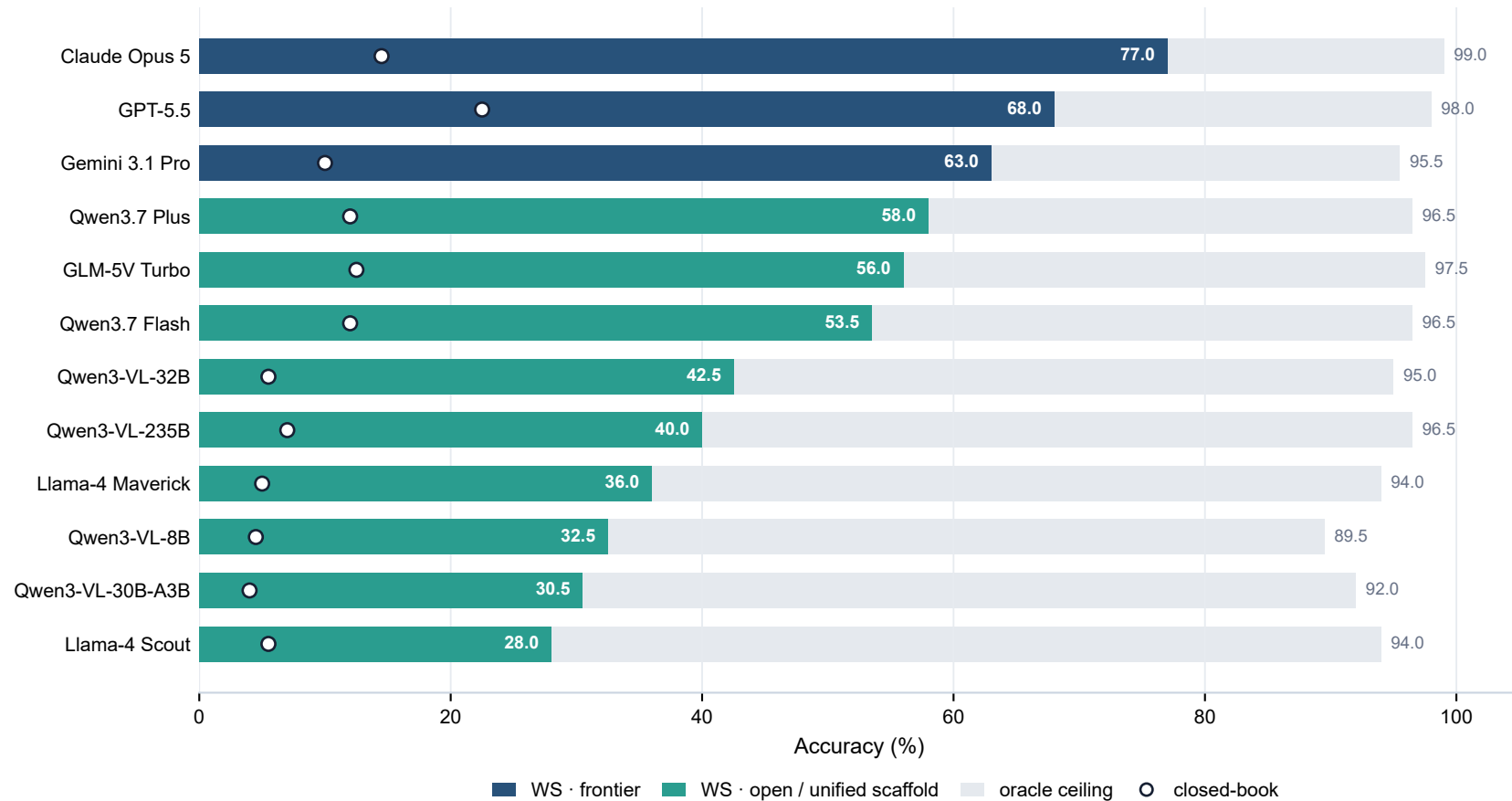


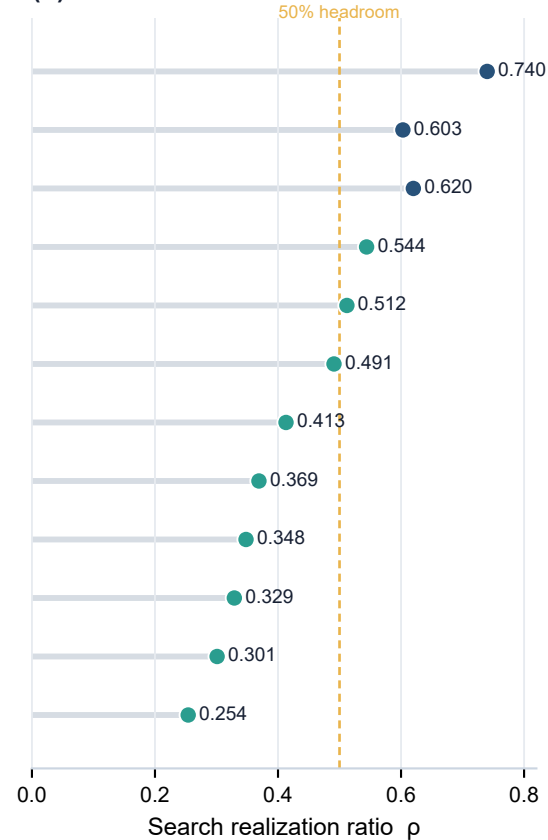
Search—not answerability—is the dominant bottleneck

Independent replication split · n = 200 · same retrieval scaffold

(a) Closed-book / with-search / oracle



(b) Certified headroom realized



Oracle remains 89.5–99.0%, while with-search spans 28.0–77.0%.